

Present-State Resolution and Lineage History in Early Mouse Embryogenesis

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Abstract

Developmental-history claims depend on how completely the contemporaneous state has been represented. We tested this dependence in 3,704 lineage-qualified cells from three E7.5 mouse embryos in the MELA dataset, joining source transcriptomes exactly to source-provided reconstructed lineage trees. The analysis separates three questions: whether coarse-state discordance survives molecular refinement, whether the molecular present predicts a disjoint molecular target beyond coarse annotations, and whether the reconstructed lineage adds predictive information beyond that present state.

A coarse state defined by source germ layer, cell-cycle phase and donor/host type contains 780,783 unordered pairs with matched coarse state, different summary of reconstructed lineages, and different contemporaneous `cells_type`. The audit of the Source-method shows that `cell_type` is expression-derived. At molecular partition size $K = 32$, 3.1338% of these pairs remain co-clustered, but an annotation-discordant control that does not require lineage discordance survives at 3.1353%. A tighter annotation-pair-matched comparison yields no corrected history-specific preference ($p = 0.527$). The raw witness collapse therefore measures resolution of expression-associated annotation discordance rather than independent evidence for lineage-specific memory. A separate diagnostic using a disjoint contemporaneous molecular target panel shows preferential separation under its matched control, while remaining a same-time molecular calibration rather than an independent phenotype.

The principal predictive analysis is fully inductive. For each held-out embryo, gene eligibility, panel construction, normalization, representation fitting, and hyperparameter selection use only the other two embryos. Predictor and target panels are disjoint and use separate library-size denominators. A fixed 64-dimensional signed-hash molecular representation improves on both the training-global-mean and S_0 -only baselines in each held-out embryo. Relative NSE reductions over S_0 are 1.87%, 2.33%, and 2.96%. The result is retained across all tested 64-dimensional state-hash seeds, larger hash dimensions, target-hash seeds, and a training-fitted state PCA32 representation; an aggressive 32-dimensional signed-hash state representation loses slightly to S_0 in two embryos. A separately fitted target-PCA32 analysis also favors molecular prediction in each embryo, with within-frame reductions of 32.37%, 17.40%, and 26.81%. Thus the molecular-over- S_0 ordering is observed under both signed-hash and conventional PCA target representations.

Lineage models do not establish an additional family-wise predictive contribution. One prospectively specified joint-metric configuration produces positive finite gains in all three embryos, with mean NSE gain 6.42×10^{-6} , but the complete 76-comparison family, including nested retuning under 999 conditional history permutations, yields $p = 0.553$. No fold reaches the prespecified 10^{-4} materiality threshold. The separate 125-bank discrete history diagnostic likewise has zero multiplicity-corrected departures (family $p = 0.31008$). No tested finite partition is promoted as biological identity, and the reconstructed internal lineage nodes have no directly measured transcriptomes. The data therefore support additional information in the measured molecular present while leaving exact state sufficiency, causal developmental memory, Markovianity, and future-state transition laws unresolved.

1 Introduction

Embryonic development couples changing molecular state with lineage history, tissue organization, and progressive restriction of developmental possibilities. Joint lineage-transcriptomic technologies increasingly measure several of these components in the same experimental system, allowing a direct empirical question: when cells differ despite sharing a coarse description of their present state, how much of that discordance remains after the present is measured more completely, and how much additional predictive information is carried by lineage history?¹

The central inferential problem is one of informational ownership. Coarse annotations, reconstructed lineage, and molecular measurements can all correlate with cellular differences. Their coexistence does not identify which representation actually carries the information needed for prediction. A stronger claim requires a controlled comparison: refine the contemporaneous state, hold out an outcome representation, and ask whether lineage contributes additional information after the refined present has been given a fair opportunity to predict.

This distinction matters especially in lineage-resolved single-cell data. A coarse state can place molecularly distinct cells in one observational category. Reconstructed lineage can separate some of those cells further. A contemporaneous molecular representation can also resolve the same differences. Demonstrating lineage distinctness therefore establishes historical structure, while the incremental predictive question asks whether lineage is required once the present molecular state is represented at higher resolution.

We address these questions using the E7.5 MELA lineage-transcriptomic resource.² The analysis uses 3,704 source-cell-ID-matched cells from three embryos. Each selected cell carries contemporaneous transcriptomic measurements and a position in a source-provided reconstructed lineage tree. Source-method audit establishes that deposited `cell_type`, germ-layer, and cell-cycle labels are expression-derived or downstream of expression-derived annotation procedures, whereas source `type` records donor/host origin. All 3,704 selected cells are donor cells, so `type` contributes no variation within the analyzed bank.

The study asks three questions:

1. Does coarse-state discordance show a history-specific molecular-resolution effect after matched controls?
2. Does a fully held-out contemporaneous molecular state predict a disjoint same-time molecular target beyond the coarse annotation state?
3. Do reconstructed-lineage summaries provide a robust incremental prediction beyond the contemporaneous molecular state?

The results define a representation-aware boundary. The historical annotation witness does not isolate a history-specific molecular-resolution effect. The fully inductive molecular present improves

¹Representative lineage-state studies include Weinreb et al. (2020), Li et al. (2023), Jia et al. (2026), and the MELA source study used here. Full reference notes appear in Appendix A.

²Appendix A, Ref. R1.

prediction over the coarse comparator in each of the three held-out embryos, under both signed-hash and training-fitted PCA target families. Representation capacity is of note: aggressive signed-hash compression can remove part of the increment. The available lineage summaries produce no family-wise predictive contribution in the tested model family. These findings identify additional information in the measured present without promoting any tested molecular representation to exact state identity or treating lineage as biologically irrelevant.

1.1 Contributions

This study makes four contributions.

1. *A matched-control analysis of coarse-state discordance.* We enumerate the complete same- S_0 pair universe, quantify cell coverage and fiber concentration, and compare molecular co-clustering survival against prespecified controls. This separates expression-derived annotation calibration from history-specific interpretation.
2. *A fully inductive cross-embryo molecular prediction experiment.* For every held-out embryo, gene eligibility, panel definition, normalization, representation fitting, and hyperparameter selection are owned by the training embryos. Molecular state is compared directly with both a global training mean and an S_0 -only baseline.
3. *A family-calibrated test of incremental lineage information.* We quantify exposure of the original exact-key history estimator, add regularized and joint-metric lineage models, retain every tested predictor/target configuration, and calibrate the complete history-model family with nested retuning under conditional permutation.
4. *A representation and diagnostic robustness surface.* We retain all tested state-hash seeds, dimensions, target-hash seeds, training-fitted PCA representations on both predictor and target sides, all 125 state/target/history ambiguity banks, simultaneous conditional permutation controls, and finite-state qualification sensitivity.

2 Results

2.1 A lineage-qualified three-embryo population with explicit annotation provenance

We analyzed the E7.5-R1, E7.5-R2, and E7.5-R3 MELA source files. The raw files contain 1,342, 1,838, and 1,468 cells, respectively. Exact source-cell-ID matching to reconstructed lineage-tree leaves yields

$$1032, \quad 1298, \quad 1374$$

lineage-qualified cells, for a total of

$$\boxed{3704}.$$

All selected cells match their source identifiers and embryo assignments exactly, with zero unmatched cells.

The source feature axis contains 78,258 unique stable gene identifiers. The predictive analysis uses fold-local gene eligibility, so the exact eligible vocabulary is allowed to differ across whole-embryo holdouts. Training on R2+R3 yields 19,412 eligible genes, divided deterministically into 13,549 state-panel genes and 5,863 target-panel genes. Training on R1+R3 yields 20,446 eligible genes, divided into 14,262 state genes and 6,184 target genes. Training on R1+R2 yields 19,432 eligible genes, divided into 13,569 state genes and 5,863 target genes. In every fold the panels are exactly disjoint.

The source-method audit also changes the interpretation of the coarse annotation state. Source **type** is donor/host origin determined from lineage-tracing-character and transcriptomic SNP evidence. All 3,704 analyzed cells are donor cells. Source **cell_type** is derived from expression-based label transfer, marker-guided manual curation, automated annotation, and grouping. Germ-layer annotation is assigned downstream of transcriptional subtype annotation, and cell-cycle phase is derived from expression-based S and G2/M scores. These variables remain valid descriptive annotations, while their provenance is of note when they are used to define a molecular-independence claim.

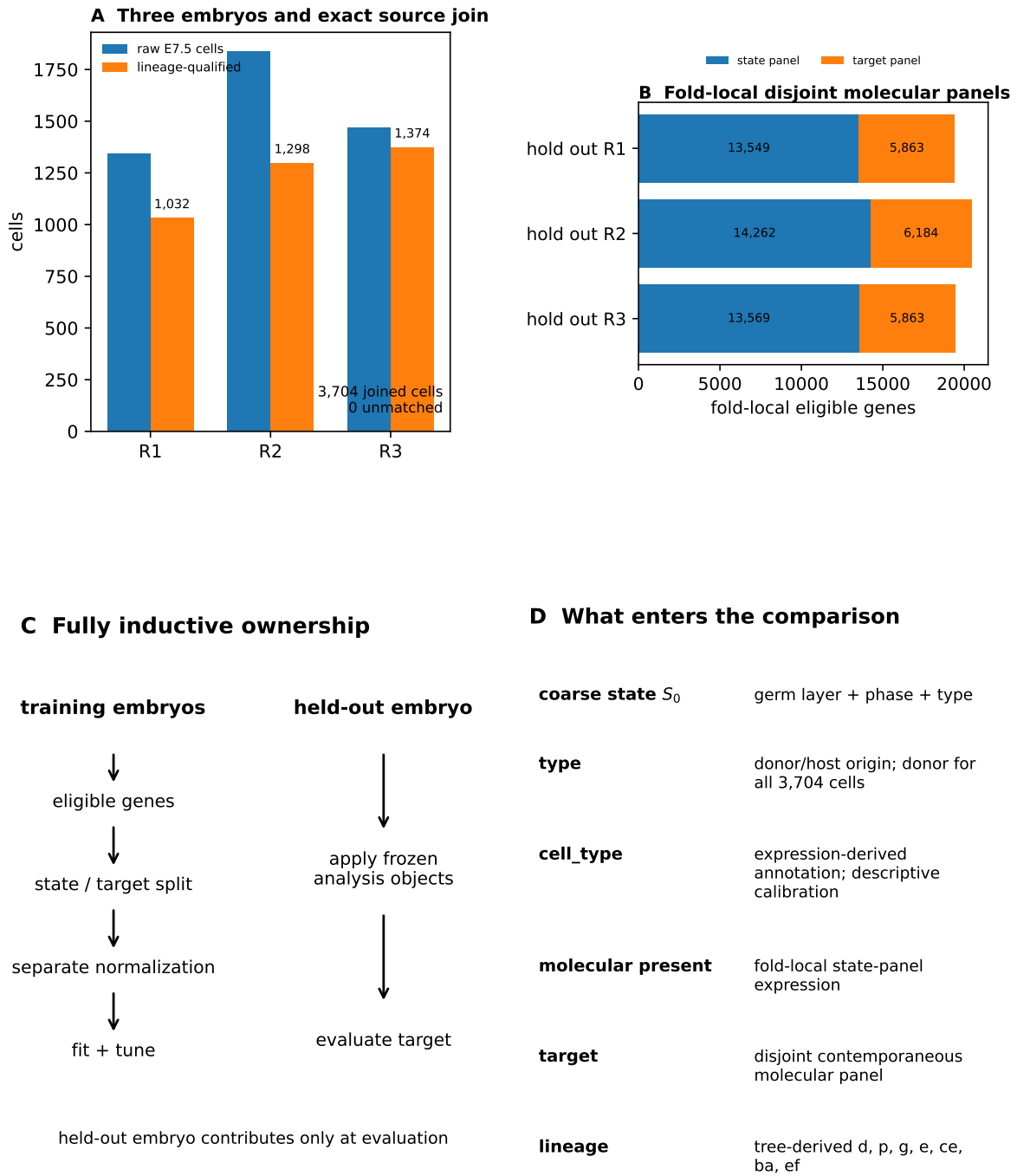


Figure 1: *Measured objects and analysis ownership.* Every fully inductive predictive fold derives gene eligibility, molecular panels, normalization, representation fitting, and model selection from the two training embryos. The held-out embryo enters only after these objects are frozen. The target panel is contemporaneous and disjoint from the state panel; it is not a future developmental state.

2.2 The historical coarse discordance bank is broad across cells and concentrated across fibers

The historical coarse state is

$$S_0(x) = (\text{germ layer}(x), \text{cell-cycle phase}(x), \text{type}(x)). \quad (1)$$

Because all selected cells are donor cells, the `type` coordinate is constant in this population.

For distinct cells x and y , the historical discordant-pair bank is defined by

$$S_0(x) = S_0(y), \quad L_4(x) \neq L_4(y), \quad A(x) \neq A(y), \quad (2)$$

where A is source `cell_type`. There are

$$\boxed{780,783}$$

such unordered pairs among

$$1,556,131$$

possible same- S_0 unordered pairs, corresponding to 50.17% of the same- S_0 pair universe.

The bank is broad at the cell level: 2,783 of 3,704 cells, or 75.13%, participate in at least one historical discordant pair. Of the 780,783 pairs, 257,489 are within embryo and 523,294 are cross-embryo. The pair count is also strongly concentrated by coarse-state fiber. The largest coarse fiber contributes 76.35% of all historical discordant pairs, and the two largest fibers contribute 96.07%.

These quantities distinguish biological coverage from combinatorial pair multiplicity. The biological replicate count remains three embryos throughout the analysis.

2.3 Matched controls identify annotation calibration rather than history-specific witness collapse

Historical molecular codes were constructed from the contemporaneous state representation at

$$K \in \{8, 16, 32, 64, 128\}.$$

Each (S_0, q_K) is a refinement of S_0 . The five independently fitted K-means solutions are not treated as a nested refinement chain.

For the historical discordant-pair family W_0 , survival at resolution K is

$$s_K^W = P(q_K(x) = q_K(y) \mid \{x, y\} \in W_0). \quad (3)$$

The historical survival curve is

$$13.60\%, \quad 6.29\%, \quad 3.13\%, \quad 2.21\%, \quad 1.26\%$$

for $K = 8, 16, 32, 64, 128$, respectively. At $K = 32$, 24,468 of 780,783 historical pairs survive, giving the previously reported 96.87% raw collapse.

Matched-control analysis changes the interpretation of this number. Let C_0 denote all same- S_0 pairs, C_1 same- S_0 annotation-discordant pairs without a lineage-discordance requirement, C_2 same- S_0 lineage-discordant pairs without an annotation-discordance requirement, C_3 the frozen broad non-witness matched control, and C_{3A} the tighter annotation-pair-matched control.

At $K = 32$,

$$s_{32}^W = 0.0313378,$$

while

$$s_{32}^{C_1} = 0.0313525.$$

Thus

$$\frac{s_{32}^W}{s_{32}^{C_1}} = 0.99953. \tag{4}$$

The annotation-discordant control therefore reproduces essentially the same molecular co-clustering survival without requiring lineage discordance.

Against the broader C_3 control,

$$s_{32}^{C_3} = 0.217437, \quad \frac{s_{32}^W}{s_{32}^{C_3}} = 0.14412.$$

This broad contrast corresponds to an 18.61 percentage-point excess collapse relative to C_3 . Under tighter annotation-pair matching, however,

$$s_{32}^W = 0.031579, \quad s_{32}^{C_{3A}} = 0.041587,$$

with corrected

$$p = 0.527.$$

No tested K shows a corrected annotation-pair-matched preference.

The source provenance and matched-control results therefore support a precise interpretation: the historical witness curve is a strong calibration of expression-derived annotation separation under finer molecular state, while a control-robust history-specific resolution effect is not established.

A separate diagnostic uses disjoint contemporaneous target-panel molecular codes in place of `cell.type`. That bank contains 1,464,732 pairs. At $K = 32$, its witness survival is 11.40% versus 28.02% for the broad matched control and 16.36% for the tighter matched control. The tighter conditional comparison yields corrected $p = 0.00775$. This result shows preferential separation of a disjoint same-time molecular target under the tested conditional null. Because both panels are contemporaneous measurements from the same cells, the result is interpreted as descriptive molecular calibration rather than an independent phenotype or future-state validation.

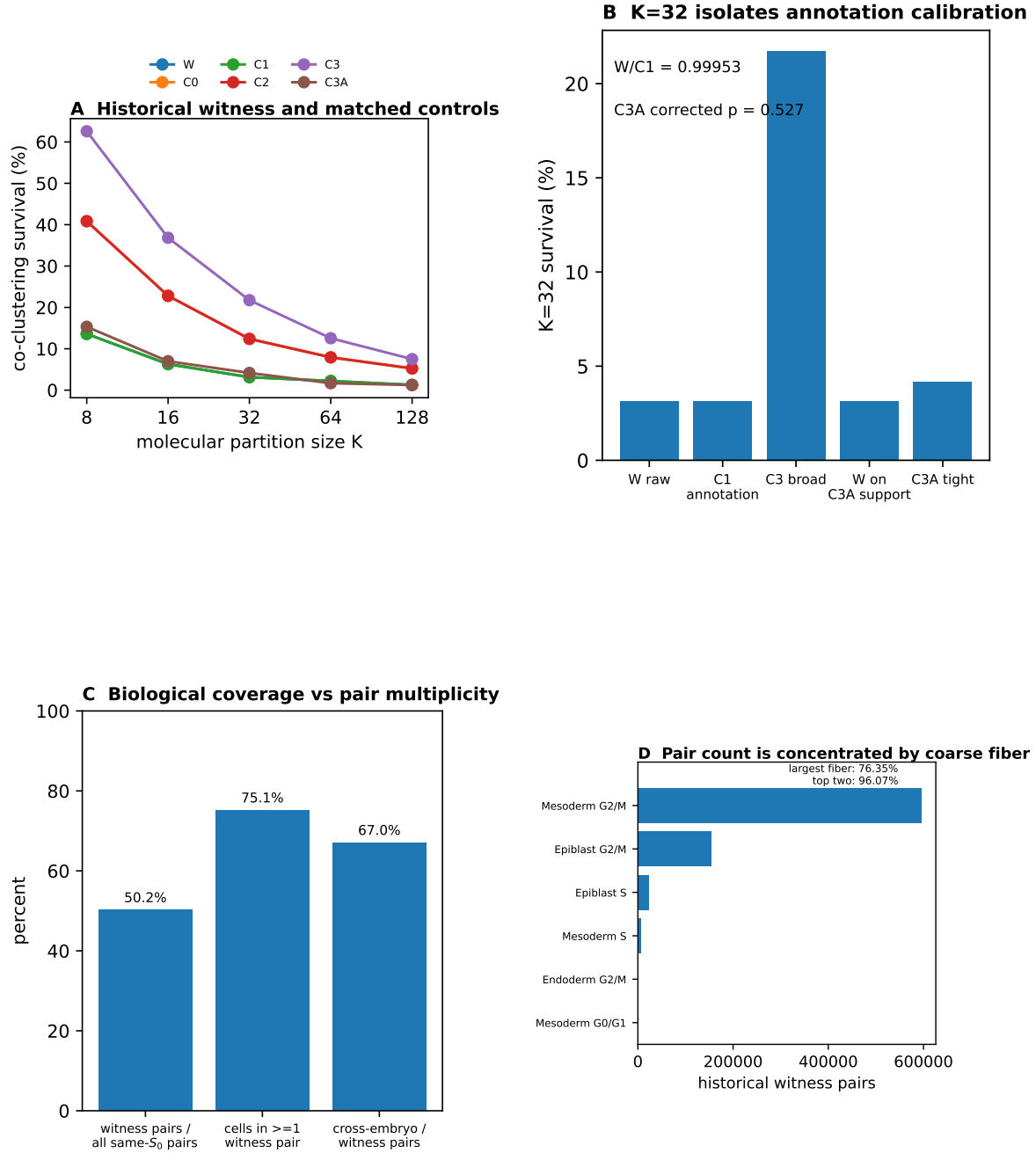


Figure 2: *Matched controls separate annotation calibration from history-specific interpretation.* Historical witness survival is nearly identical to annotation-discordant control survival. Broad non-witness matching produces a large contrast, while tighter annotation-pair matching does not produce a corrected historical preference. Pair counts are reported together with biological coverage and fiber concentration.

2.4 A fully inductive molecular present predicts a disjoint target beyond coarse annotations

The principal predictive analysis is fully inductive at the embryo level. For held-out embryo e , the other two embryos define the eligible-gene vocabulary, disjoint state and target panels, panel-specific normalization, molecular representation, and all tuned model hyperparameters. The held-out embryo contributes no information to feature eligibility or representation fitting.

For each panel $P \in \{S, T\}$, panel-normalized expression is

$$X_{xg}^P = \log \left(1 + \frac{10^4 c_{xg}}{\sum_{h \in G_P} c_{xh}} \right), \quad g \in G_P. \quad (5)$$

The state and target denominators are therefore disjoint by construction. The fixed publication reference maps the state panel to 64 signed-hash coordinates Z_S and the target panel to 64 signed-hash coordinates Z_T . A molecular-only k -nearest-neighbor model is tuned within the training embryos and then evaluated on the complete held-out embryo.

The canonical 64-dimensional representation improves on both baselines in all three embryos (Table 1).

Table 1: *Fully inductive canonical 64-dimensional molecular prediction.*

Held-out embryo	Molecular NSE	Global-mean NSE	S_0 -only NSE	Gain vs. global	Gain vs. S_0
R1	0.090707	0.098297	0.092433	0.007590	0.001726
R2	0.127342	0.135965	0.130386	0.008623	0.003044
R3	0.096237	0.105395	0.099177	0.009158	0.002940

Relative to the S_0 -only baseline, the foldwise NSE reductions are 1.87%, 2.33%, and 2.96%. The S_0 -only estimator requires a global-mean fallback for three R1 cells and for no cells in R2 or R3.

These results establish cross-embryo predictive information in the measured molecular present beyond the coarse source annotations. The target is contemporaneous, and the result concerns the tested target-coordinate representation rather than future developmental outcome or exact molecular reconstruction.

2.5 The molecular increment is broad across tested representations and sensitive to aggressive signed-hash compression

The robustness bank contains 19 prespecified predictor/target configurations rather than a full cross-product: one canonical state-hash64/canonical-target-hash64 configuration, seven additional state-hash64 seeds with the canonical target fixed, three additional state dimensions $D \in \{32, 128, 256\}$, one training-fitted state PCA32 configuration, and seven alternative target-hash64 seeds with the canonical state fixed. Thus

$$1 + 7 + 3 + 1 + 7 = 19.$$

Every configuration beats its own training-global-mean baseline in every held-out embryo. Configurations with different target coordinate systems are evaluated only against matched baselines in the same target frame; their absolute NSE values are not ranked across frames.

The stricter molecular-over- S_0 comparison is not universal. The only failed baseline gates occur for the 32-dimensional signed-hash state representation in R1 and R2:

$$R1 : 0.093546 - 0.092433 = +0.001113,$$

$$R2 : 0.130714 - 0.130386 = +0.000328.$$

In R3 the same representation remains slightly better than S_0 .

A training-fitted PCA32 representation using the same fold-local state panel improves over S_0 in all three embryos:

	$R1$	$R2$	$R3$
NSE_{PCA32}	0.084848	0.119333	0.090087
NSE_{S_0}	0.092433	0.130386	0.099177

The tested 64-dimensional state-hash seeds, larger hash dimensions, and target-seed configurations also preserve molecular improvement over S_0 in all three embryos.

The controlled failure at hash32 identifies a capacity effect rather than a simple dimensional threshold. A 32-dimensional random-feature compression can lose part of the incremental molecular geometry even when a 32-dimensional training-fitted PCA frame preserves it.

2.6 The molecular increment extends to a training-fitted non-hash target representation

The state-side PCA analysis still predicted hashed target coordinates. We therefore tested a separate training-fitted target-PCA32 representation. For each outer fold, target eligibility and target-panel normalization are owned by the training embryos; the target mean and PCA loadings are fitted only on training cells and applied unchanged to the held-out embryo. The same exclusion is repeated inside inner tuning, and neighborhood size is retuned for the PCA target frame.

Table 2: *Training-fitted PCA32 target prediction.* Each row compares models in one outer-training-owned target frame. Reduction is relative to that row’s S_0 -only baseline.

Embryo	k	Molecular NSE	S_0 NSE	Global NSE	Gain vs. S_0	Reduction (%)
R1	50	0.574233	0.849102	1.871980	0.274870	32.37
R2	25	0.745706	0.902760	2.066584	0.157053	17.40
R3	25	0.588839	0.804507	1.732313	0.215668	26.81

Molecular prediction improves on both the global-mean and S_0 -only comparators in all three PCA-target folds. A fixed unnormalized-PC sensitivity analysis gives the same ordering. The molecular-over- S_0 result is therefore observed under both signed-hash and training-fitted PCA target families. This establishes robustness across the tested target representation families without implying representation independence.

2.7 The historical exact-key lineage estimator had little exposure at deeper history levels

The historical lineage estimator retained a fixed molecular neighborhood and changed only the prediction weights of neighbors with an exactly matching discrete history key. Its capacity therefore depends on how often such matches are present among molecular neighbors.

For each held-out cell and history level L_h , let

$$m_h(x) = \#\{y \in N_{25}(x) : L_h(y) = L_h(x)\}.$$

At L_1 , zero-match fractions are 32.36%, 23.19%, and 30.64% for R1, R2, and R3. At L_2 , the corresponding fractions are 88.86%, 87.67%, and 88.79%. At deeper summaries they approach complete non-exposure:

	<i>R1</i>	<i>R2</i>	<i>R3</i>
L_3 zero-match fraction	97.29%	97.23%	97.23%
L_4 zero-match fraction	97.67%	97.61%	97.89%

The normalized prediction weights are therefore unchanged for the same fractions of cells. Mean total-variation changes in the prediction weights are approximately 0.051–0.061 for L_1 , 0.0047–0.0051 for L_2 , about 0.0010 for L_3 , and below 0.001 for L_4 .

The historical L_3 and L_4 NSE gains are numerically stable under float64 recomputation, with discrepancies below 2.1×10^{-13} , but they remain positive in only two of three embryos and have magnitudes near 10^{-6} . Their principal limitation is therefore estimator capacity and biological magnitude rather than floating-point instability.

2.8 Lineage models show no family-wise predictive contribution in the tested family

We next tested lineage in models where history can alter prediction independently of an already selected molecular neighborhood. Seven continuous tree-derived variables are used:

$$H = (d, p, g, e, ce, ba, ef),$$

representing depth, parent arity, grandparent arity, incoming edits, cumulative edits, branching-ancestor count, and observed leaf edit fraction. Raw clone identity, node identity, and path identity are excluded.

Two model families are evaluated under training-only inner tuning. The first compares molecular-only ridge prediction with ridge prediction using molecular state plus standardized history variables. The second modifies the neighbor metric:

$$D_\lambda(x, y) = D_S(x, y) + \lambda \frac{D_H(x, y)}{7}, \tag{6}$$

where D_S is squared molecular distance and D_H is squared distance in standardized transformed history coordinates.

The complete predictive lineage family contains both original and prospectively extended tuning grids, both history-model families, and all 19 predictor/target configurations. This yields

$$19 \times 2 \times 2 = 76$$

three-embryo comparison units, comprising 228 outer-fold lineage comparisons. No unfavorable configuration is omitted. One extended joint-metric configuration, state `hash64_seed6`, produces positive finite NSE gains in all three embryos. Its independently selected parameters and foldwise effects are:

Table 3: *The sole three-embryo positive lineage configuration.* Molecular and history-augmented neighborhood sizes are tuned independently within the training embryos.

Embryo	k_M	k_H	λ	Molecular NSE	Molecular+history NSE	Absolute gain	Reduction (%)
R1	50	50	0.001	0.091098990	0.091095046	3.944e-06	0.00433
R2	50	50	0.0001	0.127926622	0.127921104	5.518e-06	0.00431
R3	50	50	0.001	0.096717440	0.096707638	9.801e-06	0.01013

The mean finite gain is

$$\bar{G} = 6.420989 \times 10^{-6},$$

and no fold reaches the prespecified absolute materiality threshold 10^{-4} .

To account for the complete search family, define for member m

$$T_m = \min_{e \in \{R1, R2, R3\}} G_{m,e}, \quad T_{\max} = \max_{m=1, \dots, 76} T_m.$$

A member has positive T_m only if its lineage-augmented model improves all three embryos. In each of 999 fixed null realizations, whole seven-feature history tuples are reassigned within embryo $\times$ historical $S_0 \times$ target-blind molecular-neighborhood strata. All history-bearing inner tuning grids are rerun before outer evaluation. Molecular predictors, target representations, and embryo ownership remain fixed.

The observed family maximum is

$$T_{\max}^{\text{obs}} = 3.944081744 \times 10^{-6}.$$

Among 999 null realizations, 552 maxima are at least this large, giving the inclusive plus-one family probability

$$p_{\text{family}} = \frac{1 + 552}{1000} = 0.553.$$

The isolated three-embryo positive configuration therefore does not retain family-wise support. Within the tested model family, the data do not establish an incremental lineage contribution beyond the contemporaneous molecular predictor. This conditional calibration assumes exchangeability of history tuples within the declared finite strata; it does not establish causal independence or population-level absence of lineage effects.

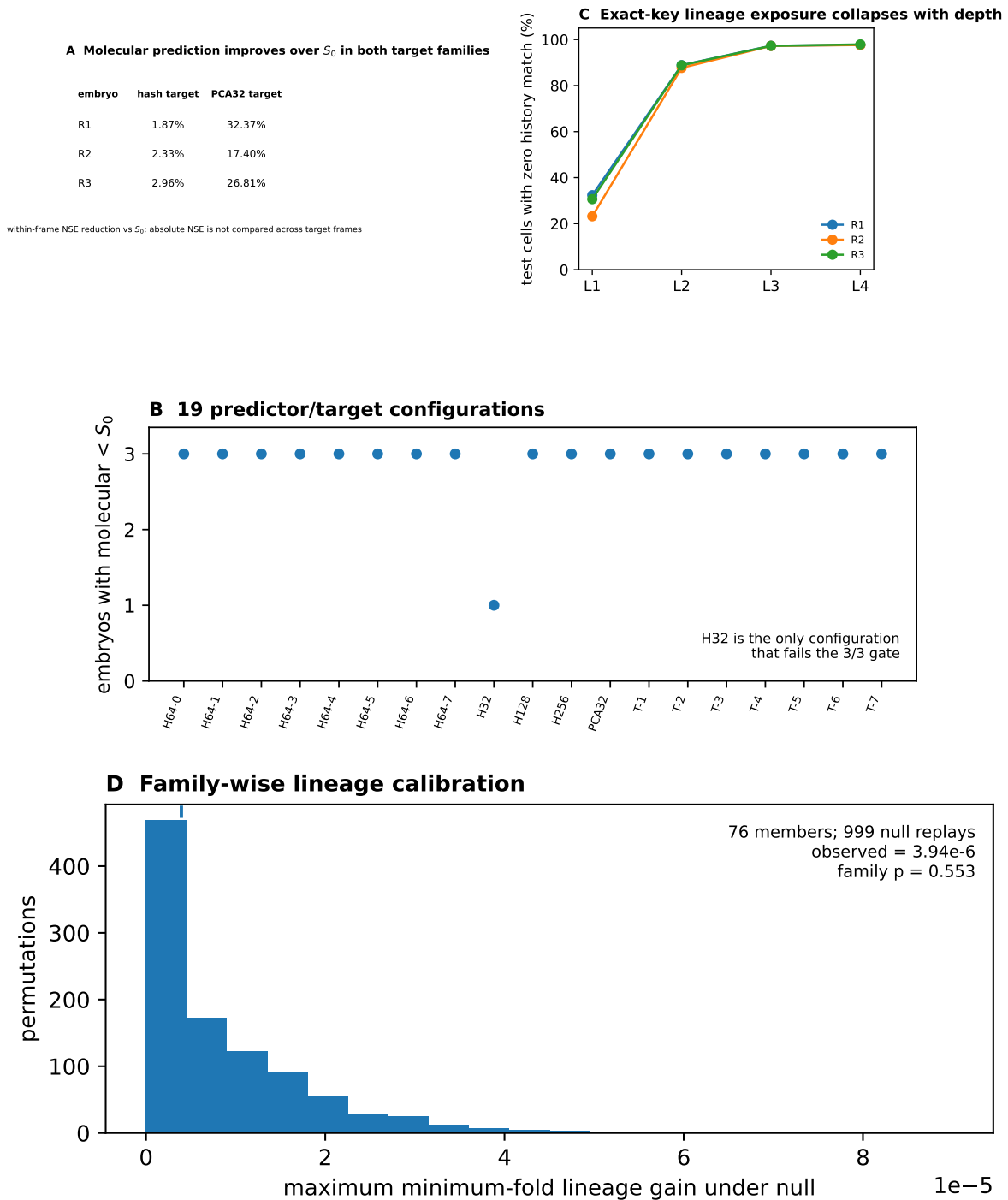


Figure 3: *Molecular prediction and lineage calibration are evaluated separately.* Molecular state improves on the coarse comparator in each held-out embryo under both hashed and PCA target families. The isolated positive lineage configuration is small and does not survive complete-family calibration.

2.9 The complete ambiguity/support family has no multiplicity-corrected departure

A separate discrete diagnostic groups cells by a state code and optional history key and measures disagreement in a contemporaneous target code. For key-defined fibers F , let

$$P = \sum_F \binom{|F|}{2}$$

be the total number of within-key unordered collision pairs and

$$D = \sum_F \left[\binom{|F|}{2} - \sum_z \binom{n_{F,z}}{2} \right]$$

be the number with different target codes. Ambiguity is

$$A = \frac{D}{P}.$$

All 25 state-code/target-code combinations and all five history levels L_0 – L_4 are retained, producing 125 diagnostic banks. Richer history summaries frequently lower observed ambiguity, while also reducing collision support sharply. The originally highlighted K32/K64 bank is one member of this complete surface and is not privileged prospectively.

A coherent conditional permutation family uses common strata defined by S_0 , embryo, and the joint five-state-code vector, with a simultaneous maximum statistic across the 100 history contrasts. The movable-cell fraction is 0.49784. The observed family maximum is 0.12556, with

$$p_{\text{family}} = 0.31008$$

and

$$0$$

corrected departures.

This negative family-level result replaces any interpretation of an isolated nominal K32/K64 departure as a robust history effect. It does not constitute a powered population-level absence test; it establishes the outcome of the complete finite conditional diagnostic family used here.

2.10 Finite molecular partitions remain diagnostic rather than biological identities

The historical finite-state analysis evaluated $K = 8, 16, 32, 64, 128$ against recurrence and non-fingerprint criteria. None of the five historical partitions passes all four prespecified gates. A broader threshold-sensitivity grid evaluates 405 combinations of singleton fraction, median recurrent-fiber size, recurrent-fiber count, and largest-fiber fraction. Forty-five combinations pass their numerical diagnostic thresholds.

The sensitivity result clarifies the role of these gates. They describe recurrence and concentration properties of finite partitions; they do not certify biological identity. In particular, the historical K32 value of 99 recurrent fibers relative to a threshold of 100 is not treated as a biologically meaningful near miss.

No tested finite partition is therefore promoted as an exact state quotient.

2.11 The available lineage trees do not identify a molecular continuation law

The source lineage trees contain 2,686 reconstructed internal nodes. None of these internal nodes has a directly measured transcriptome:

$0/2686$.

A molecular continuation experiment requires a measured ancestral molecular state, a keyed lineage relation, and a measured descendant outcome. The present source provides reconstructed topology and terminal-cell transcriptomes without measured ancestral transcriptomes. The current data therefore do not identify an ancestral-state transition distribution.

The empirical authority of this study ends at contemporaneous representation, cross-embryo same-time molecular prediction, and finite tests of incremental information from reconstructed-lineage summaries. Exact state sufficiency, causal developmental memory, Markovianity, non-Markovianity, and future-fate prediction remain outside this design.

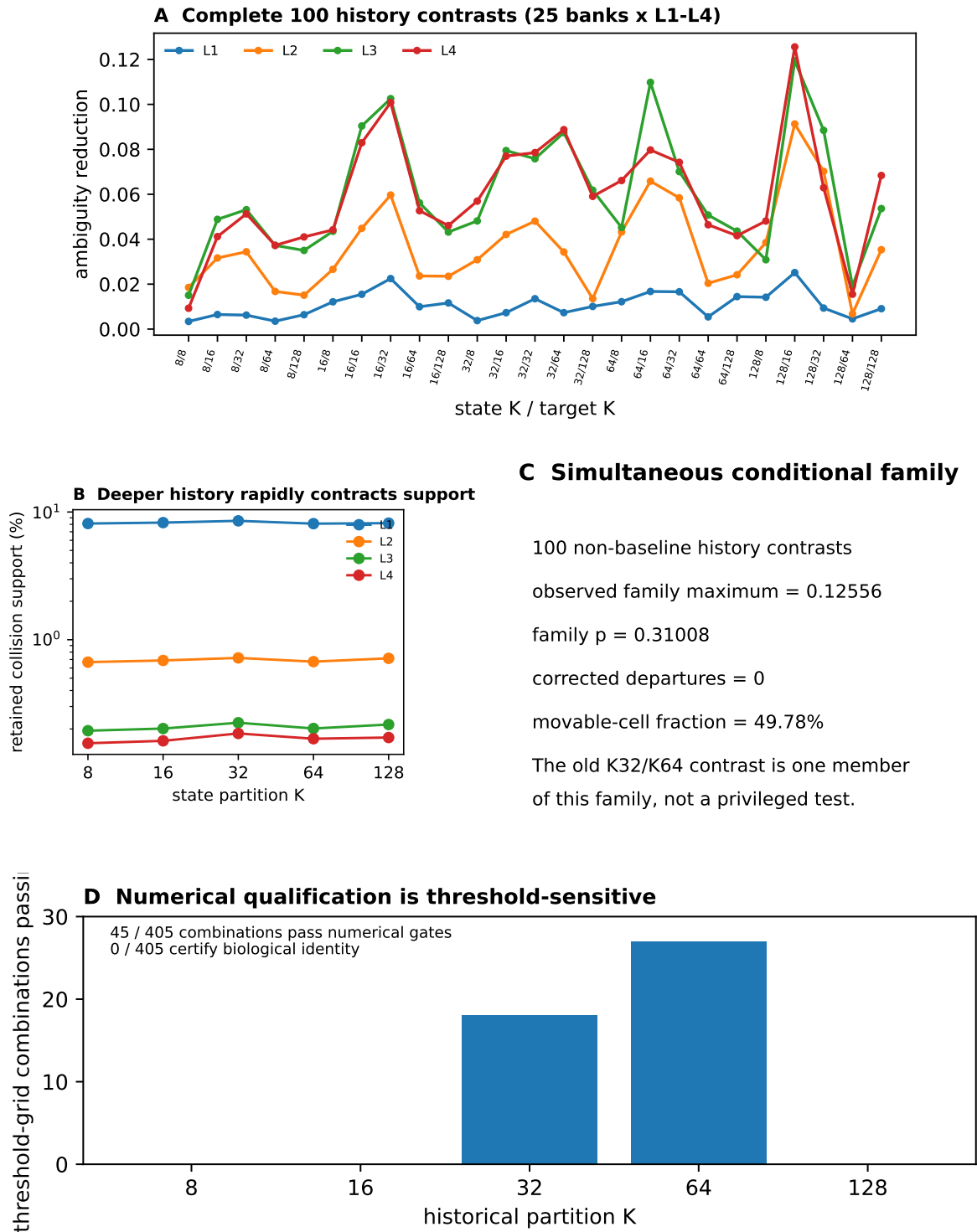


Figure 4: Complete discrete diagnostics preserve negative family-level results and state-identity boundaries. All diagnostic banks are shown together. Lower ambiguity under richer conditioning is interpreted with retained support, multiplicity control, and the finite nature of the tested partition family.

3 Discussion

Five conclusions emerge from these analyses. Together they support a representation-first interpretation: association with a biological variable does not by itself establish that the variable owns the information needed to explain or predict the observed difference.

3.1 Coarse annotation discordance does not isolate a lineage-specific memory effect

The selected population contains extensive discordance inside the coarse source annotation state. More than half of all same- S_0 pairs satisfy the historical combination of different L_4 history and different `cell_type`, and more than three quarters of analyzed cells participate in at least one such pair.

Source provenance and matched controls specify the meaning of this result. `cell_type` is expression-derived, and its molecular co-clustering survival is essentially reproduced by annotation-discordant pairs without a lineage-discordance requirement. The 96.87% K32 collapse therefore measures how strongly a finer molecular representation resolves expression-associated annotation discordance. It does not isolate a history-specific molecular-resolution effect.

This leaves two positive facts intact: the coarse annotation state is incomplete for contemporaneous molecular organization, and genuine reconstructed-lineage differences exist within those coarse fibers. Their coexistence does not assign the molecular difference to lineage.

3.2 The measured molecular present adds predictive information beyond the coarse comparator in each tested embryo

The clearest positive result is the fully inductive cross-embryo prediction experiment. The canonical 64-dimensional molecular representation improves on both the global training mean and the S_0 -only predictor in R1, R2, and R3. Relative NSE reductions over S_0 are 1.87%, 2.33%, and 2.96%. The effect is modest and consistent in direction across the complete set of three biological embryos.

The same ordering persists when the target representation is changed from signed hashing to training-fitted PCA32. In those PCA target frames, molecular-over- S_0 reductions are 32.37%, 17.40%, and 26.81%. These percentages belong to a different outcome geometry and are not directly comparable in magnitude to the hash-target percentages. Their significance is qualitative: the molecular-over- S_0 ordering is not confined to one target hash family.

The result establishes additional information in the measured contemporaneous molecular state under strict holdout. It does not establish that the measured molecular state is sufficient, uniquely privileged, or equivalent to a future developmental state.

3.3 Representation capacity determines how much of that information is retained

The robustness bank contains a controlled failure. A 32-dimensional signed-hash state representation loses slightly to the S_0 -only baseline in two embryos, despite beating the global mean in all three. PCA32 succeeds on the same molecular panel, and 64-, 128-, and 256-dimensional signed-hash representations retain the molecular increment.

This result identifies representation capacity as part of the empirical object. The measured signal is not tied to one favorable random seed, while aggressive random-feature compression can discard information that a training-fitted representation of the same nominal dimension preserves. The result therefore supports a family of informative present-state representations rather than a claim that one coordinate construction is the biological state itself.

3.4 The tested lineage family does not earn an additional predictive contribution

The original exact-key estimator had little opportunity to act at deeper history levels: approximately 97–98% of held-out cells had no L_3 or L_4 match among their molecular neighbors. Its near-zero gains therefore do not support a general claim that lineage is irrelevant.

The broader model family supplies the stronger test. One prospectively specified joint metric gives positive finite gains in all three embryos, but the gains are extremely small and the complete 76-member family calibration yields $p = 0.553$. The finite positive configuration is retained as an observed result; it does not provide family-wise evidence for an incremental lineage contribution.

The appropriate conclusion is therefore bounded. In the tested contemporaneous prediction problem, the molecular present improves prediction over coarse annotations in each of the three held-out embryos, whereas the available lineage summaries do not earn a robust additional share after multiplicity and model-selection opportunity are accounted for. This does not exclude lineage-dependent fate bias, heritable epigenetic structure, or other historical information under different targets and experimental designs.

3.5 Finite state and transition claims remain beyond the present data

The complete 125-bank ambiguity/support family produces no multiplicity-corrected history departure. Finite-state qualification is threshold-sensitive and does not certify biological identity. The reconstructed lineage trees contain 2,686 internal nodes and no directly measured internal-node transcriptomes.

These results place a precise endpoint on the paper. The data identify additional contemporaneous molecular information and quantify the tested lineage contribution. They do not identify an exact state equivalence relation, a causal memory variable, a Markov or non-Markov transition law, or an ancestor-to-descendant molecular continuation map. Those unresolved objects motivate analyses based on local relational geometry rather than promotion of a finite persistent state.

3.6 Relation to lineage and trajectory approaches

Trajectory inference and optimal-transport methods reconstruct developmental order or coupling from sampled molecular states.³ Lineage tracing adds experimentally recorded or computationally reconstructed ancestry, enabling different questions about inherited structure and fate. Studies linking lineage to later fate have shown predictive value for ancestry in temporal designs, and multimodal lineage systems extend this information to epigenomic and spatial measurements.⁴

The present analysis addresses a same-time molecular target. Its lineage question is explicitly incremental: after constructing a held-out representation of the present, do the available reconstructed-lineage summaries improve prediction of a disjoint contemporaneous molecular panel? The family-calibrated answer reported here is compatible with stronger lineage effects on future fate or other modalities.

The absence of directly measured internal-node transcriptomes also separates this analysis from process-level Markov tests. Mathematical lumpability conditions concern transition distributions of a stochastic process.⁵ The present data do not supply the measured ancestral molecular states needed for such a test.

4 Scope and limitations

Single developmental stage. All primary evidence comes from E7.5. The analysis does not establish a cross-stage developmental law.

Three biological embryos. The embryo is the biological replication unit. The 3,704 cells, 780,783 historical discordant pairs, 1,464,732 disjoint-target diagnostic pairs, and permutation banks do not increase the independent biological replicate count beyond three.

Selected lineage-qualified population. The 3,704-cell bank contains cells that can be matched exactly to reconstructed lineage-tree leaves. It is not the complete E7.5 source-cell population.

Reconstructed lineage authority. Lineage variables are derived from source-provided reconstructed trees. Source tree-reconstruction uncertainty is not propagated through the present analysis.

Annotation provenance. The source paper documents expression-derived construction of `cell.type` and cell-cycle annotations and downstream assignment of germ-layer labels. Exact source assignment code and complete per-label feature lists were not recovered. The historical annotation witness is therefore interpreted as descriptive annotation calibration.

³Appendix A, Refs. R3 and R4.

⁴Appendix A, Refs. R2, R5, and R6.

⁵Appendix A, Ref. R7.

Contemporaneous target. The held-out target genes are measured in the same cell and at the same developmental stage as the predictor genes. Panel disjointness and separate denominators create measurement separation; they do not create temporal or stochastic independence.

Representation dependence. The molecular-over- S_0 ordering is observed across the tested 64-dimensional state seeds, larger signed-hash dimensions, state PCA32, multiple target-hash seeds, and a training-fitted target-PCA32 family. The 32-dimensional signed-hash state representation fails the S_0 comparison in two embryos. These results establish robustness across tested representation families, not representation independence.

Lineage-model scope. The lineage analysis tests exact-key reweighting, regularized molecular-plus-history regression, and joint molecular/history distance models across 76 three-embryo comparison units. One joint-metric configuration produces a small 3/3 positive finite gain, but the complete family calibration gives $p = 0.553$. The conditional null assumes within-stratum exchangeability of history tuples, and the model family does not exhaust possible lineage representations.

Finite state qualification. No tested historical K passes the original recurrence and non-fingerprint gates, and threshold sensitivity demonstrates that numerical qualification depends on the chosen diagnostic thresholds. No finite partition is treated as biological identity.

Continuation authority. The reconstructed lineage trees contain no directly measured ancestral transcriptomes at their 2,686 internal nodes. Process-level continuation, causal-memory, and Markov claims remain unidentified.

5 Methods

5.1 Study design and analysis chronology

The study contains a preserved initial specification and a final fully inductive analysis. The initial specification generated the coarse discordant-pair bank, full-bank molecular partitions, a transductively fixed gene vocabulary, and an annotation-assisted prediction pipeline. Those computations remain unchanged and are retained in Appendix B to document why the final design uses fold-local eligibility, separate panel denominators, annotation-free molecular neighborhoods, and explicit lineage-family calibration.

The final empirical tasks are: (i) enumerate and decompose the full same- S_0 pair universe; (ii) compare pair survival with prespecified matched controls; (iii) construct a fully inductive cross-embryo molecular prediction problem with fold-local feature discovery and disjoint panel normalization; (iv) test both hashed and PCA predictor/target representations; (v) quantify exposure of the initial lineage estimator; (vi) test regularized and joint-metric lineage models under training-only tuning and complete-family conditional calibration; and (vii) evaluate the complete discrete diagnostic family without selecting favorable members after outcome inspection.

Biological replication is evaluated across the three complete embryo holdouts. Cell- and pair-level summaries and conditional permutations are finite descriptive or conditional quantities and do not increase the independent biological replicate count.

5.2 MELA source data and exact transcriptome–lineage join

We used MELA Zenodo v1, record 10.5281/zenodo.19892785, with E7.5-R1, E7.5-R2, and E7.5-R3 source files.⁶ The raw files contain 1,342, 1,838, and 1,468 cells.

For each reconstructed tree, observed source cell identifiers assigned to that tree are intersected with the tree’s node identifiers and required to admit a reconstructable path to a root. A stored cell hash is

$$\text{cellHash}(x) = \text{hex}[\text{SHA256}(\text{UTF8}(\text{sourceID}(x)))]_{0:16} . \quad (7)$$

Every join is verified against the full source identifier and replicate identity; source row order is not assumed. The resulting selected counts are 1,032, 1,298, and 1,374 cells.

5.3 Source annotation provenance

Source categorical fields are decoded directly from the deposited H5TD files and audited against the source Methods. The coarse state is

$$S_0(x) = (\text{germ_layer}(x), \text{phase}(x), \text{type}(x)) . \quad (8)$$

Source **type** denotes donor/host origin, assigned using lineage-tracing-character expression, transcriptomic SNP evidence, and doublet-exclusion criteria. All selected cells are donor cells. Source **cell_type** groups transcriptional subtypes constructed through reference expression label transfer, marker-guided manual curation, automated annotation, and grouping. Germ-layer labels are assigned to annotated transcriptional subtypes. Cell-cycle phase is based on expression-derived S and G2/M scores. Exact assignment code and some feature lists were not identified in the deposited analysis surface and are recorded as unavailable rather than inferred.

The source **cell_type** field is therefore treated as a contemporaneous expression-derived annotation. It is not a future fate variable or an expression-independent phenotype.

5.4 Reconstructed lineage variables and historical summaries

For each selected leaf, parent links are followed to the tree root. We record edge depth d , parent arity p , grandparent arity g , final incoming edit count e , cumulative root-to-leaf edit count ce , number of branching ancestors ba , and observed leaf edit fraction ef .

⁶Appendix A, Ref. R1.

The historical fixed summaries are

$$L_1 = (\min(d, 12), \min(p, 5)), \quad (9)$$

$$L_2 = (L_1, \min(g, 5), \min(e, 5)), \quad (10)$$

and

$$L_3 = (L_2, \min(\lfloor ce/5 \rfloor, 12), \min(ba, 8)), \quad (11)$$

$$L_4 = (L_3, \min(\lfloor 10ef \rfloor, 9)). \quad (12)$$

Raw clone identity, node identity, raw path identity, developmental stage, and present annotations are not used as unrestricted history predictors.

For the continuous joint models, the unbinned history vector is

$$H = (d, p, g, e, ce, ba, ef).$$

The first six variables are transformed by $\log(1 + x)$, then standardized from the relevant training data. The edit fraction is standardized without the logarithmic transform.

5.5 Coarse discordant-pair universe, matched controls, and witness permutations

The complete same- S_0 unordered pair universe is

$$C_0 = \{\{x, y\} : x \neq y, S_0(x) = S_0(y)\}.$$

The historical pair family is

$$W_0 = \{\{x, y\} \in C_0 : L_4(x) \neq L_4(y), A(x) \neq A(y)\}, \quad (13)$$

where A is contemporaneous `cell_type`.

Prespecified comparison families are: C_0 , all same- S_0 pairs; C_1 , same- S_0 annotation-discordant pairs without requiring lineage discordance; C_2 , same- S_0 lineage-discordant pairs without requiring annotation discordance; C_3 , a broad non-witness control matched on coarse stratum and unordered embryo pair; and C_{3A} , a tighter comparison that additionally matches unordered annotation pair on common support.

For molecular code q_K ,

$$s_K^B = P(q_K(x) = q_K(y) \mid \{x, y\} \in B), \quad B \in \{W, C_0, C_1, C_2, C_3, C_{3A}\}. \quad (14)$$

For a supported matched stratum s , let w_s be its number of witness pairs and let C_s be its set of non-witness complement pairs, with $c_s = |C_s|$. Each pair in C_s receives weight w_s/c_s . The

matched-control survival is therefore

$$s_K^C = \frac{\sum_{s:w_s>0, c_s>0} \frac{w_s}{c_s} \sum_{\{x,y\} \in C_s} \mathbf{1}[q_K(x) = q_K(y)]}{\sum_{s:w_s>0, c_s>0} w_s}.$$

Thus each supported matched stratum contributes total control mass equal to its witness mass. C_3 matches on S_0 and unordered embryo pair; C_{3A} additionally matches on unordered `cell_type` pair and retains 98.64% of witness support.

The witness permutation family uses 128 jointly permuted five-state-code vectors at the cell level within (S_0, embryo) . Within each stratum and permutation replicate, one random permutation of cell labels is drawn and the complete five-component molecular-code vector

$$(q_8, q_{16}, q_{32}, q_{64}, q_{128})$$

is reassigned according to that same permutation. The five resolutions therefore move jointly under a common cell relabeling. Annotation, history, witness membership, matching weights, and the dependence induced by pairs sharing cells remain fixed. For each control define

$$\delta_K = s_K^W - s_K^C,$$

and for every permutation record the maximum negative difference over $K \in \{8, 16, 32, 64, 128\}$. The corrected upper-tail probability at a given K is

$$p_{\text{corr}} = \frac{1 + \#\{b : \max_{K'}(-\delta_{K'}^{(b)}) \geq -\delta_K^{\text{obs}}\}}{129}.$$

The minimum attainable value is $1/129$. C_3 and C_{3A} use separate five- K maxima rather than a combined cross-control family. For C_{3A} at $K = 32$, 67 of 128 null maxima exceed the observed preference, giving $(1 + 67)/129 = 0.5271318$. These permutations are conditional diagnostics and are not biological replicates.

5.6 Finite molecular partitions and qualification sensitivity

The historical full-bank state coordinates are partitioned independently at

$$K \in \{8, 16, 32, 64, 128\}$$

using deterministic K-means. The clustering input is the historical 64-dimensional state coordinate matrix Z_S : counts for the frozen state-gene panel are normalized by the complete whole-cell library size,

$$\log\left(1 + 10^4 c_{xg} / L_x^{\text{whole}}\right),$$

projected with the signed-hash seed `RMM012_STATE_HASH_PROJECTION_V1`, and rowwise L2-normalized. The three embryos are stacked in the frozen 3,704-cell source order, and the stored clustering input

is float32. No additional scaling is applied before K-means.

For each K , one deterministic start is used. The first center is the row at index

$$H(\text{RMM012_FULL_STATE_KK}) \bmod 3704,$$

and subsequent centers are added by farthest-first selection: at each step the row with the largest current minimum squared Euclidean distance to the selected centers becomes the next center. Ties follow the first occurrence in the frozen row order. Lloyd updates then assign each row to its nearest center, with distance ties going to the lowest center index, and replace each nonempty center by its member mean. An empty cluster is reseeded with the row having the largest current minimum squared distance to any center. Iteration stops when assignments are unchanged and the updated centers satisfy NumPy’s default `allclose` criterion ($\text{rtol} = 10^{-5}$, $\text{atol} = 10^{-8}$), or after 80 iterations. No post-fit cluster-label canonicalization is applied.

For the discrete target diagnostics, target codes at

$$K_T \in \{8, 16, 32, 64, 128\}$$

are fitted independently to the historical 64-dimensional target coordinate matrix Z_R . Z_R is constructed from the frozen target-gene panel using the same whole-cell library normalization, the signed-hash seed `RMM012_REALIZATION_HASH_PROJECTION_V1`, and rowwise L2 normalization, with no additional scaling before clustering. Target K-means uses the same deterministic farthest-first Lloyd procedure with seed `RMM012_FULL_REALIZATION_KK`. State and target clustering are fitted separately. Each fitted (S_0, q_K) refines S_0 ; no nesting relation among the independently fitted K values is assumed.

The historical qualification gates are

$$f_1 \leq 0.25, \quad \text{median}_{F:|r(F)| \geq 2} |F| \geq 4, \quad \#\{F : |r(F)| \geq 2\} \geq 100, \quad \max_F |F|/3704 \leq 0.25, \quad (15)$$

where f_1 is the singleton-fiber fraction and $r(F)$ the set of embryos represented in fiber F . A threshold-sensitivity grid separately evaluates singleton maxima 0.20, 0.25, 0.30, median minima 3, 4, 5, recurrent-fiber minima 80, 100, 120, and largest-fiber maxima 0.20, 0.25, 0.30. Passing these numerical gates is not defined to certify biological identity.

5.7 Operational coarse state in the selected population

The state key remains

$$S_0 = (\text{germ layer}, \text{phase}, \text{type}).$$

All 3,704 selected cells have `type = donor`. Operational variation in this population is therefore carried by

$$\boxed{\text{germ layer} \times \text{cell-cycle phase}}.$$

The deposited germ-layer and phase labels remain useful coarse comparators, while their expression-derived provenance prevents treating S_0 as an expression-independent phenotype.

5.8 Fully inductive fold-local gene eligibility

For each outer held-out embryo e , all feature discovery uses only the other two embryos. A source gene is eligible when it is detected in at least 1% of cells in each of the two training embryos, has positive mean count across the training cells, and its source gene name does not begin with mitochondrial or ribosomal prefixes `mt-`, `Mt-`, `MT-`, `Rpl`, or `Rps`.

The resulting outer-fold eligible counts are 19,412 for training R2+R3, 20,446 for R1+R3, and 19,432 for R1+R2. The held-out embryo is not used to decide eligibility.

The deterministic hashing operations below use one fixed function throughout. For any character string u ,

$$H(u) = \text{uint64}_{\text{BE}}(\text{SHA256}(\text{UTF8}(u))[0:8]),$$

where SHA-256 returns a 32-byte digest, the first eight digest bytes are interpreted as one unsigned 64-bit integer in big-endian order, and the remaining digest bytes are not used by the modulo and parity operations below. String concatenation uses the literal separators shown in the equations. Hash definitions, seeds, and source gene identifiers are fixed independently of expression values and held-out outcomes.

For eligible stable gene ID g , panel assignment uses the deterministic historical split rule

$$d_g = H(\text{RMM012_STATE_REALIZATION_V1}|g) \bmod 10, \quad (16)$$

with state assignment for $d_g < 7$ and target assignment otherwise. Because eligibility is fold-local, panel counts vary by fold while the assignment rule remains fixed. The two panels are exactly disjoint.

5.9 Separate panel normalization and molecular representations

For panel $P \in \{S, T\}$, let

$$L_x^P = \sum_{g \in G_P} c_{xg}.$$

Panel-normalized expression is

$$X_{xg}^P = \log \left(1 + \frac{10^4 c_{xg}}{L_x^P} \right), \quad g \in G_P. \quad (17)$$

State and target panels therefore use separate library-size denominators.

For signed-hash seed s and dimension D ,

$$b_s(g) = H(s|\text{bucket}|g) \bmod D, \quad \sigma_s(g) = \begin{cases} +1, & H(s|\text{sign}|g) \equiv 0 \pmod{2}, \\ -1, & \text{otherwise.} \end{cases} \quad (18)$$

The projected coordinate vector is

$$h_j^P(x) = \sum_{g \in G_P: b_s(g)=j} \sigma_s(g) X_{xg}^P, \quad Z_P(x) = \frac{h^P(x)}{\|h^P(x)\|_2}. \quad (19)$$

The robustness bank contains 19 prespecified predictor/target configurations: one canonical state-hash64/canonical-target-hash64 configuration; seven additional state-hash64 seeds with the canonical target fixed; three additional canonical-seed state dimensions 32, 128, 256; one training-fitted state PCA32 configuration predicting the canonical hashed target; and seven alternative target-hash64 seeds with the canonical state fixed. This is not a state $\times$ target cross-product. Every target coordinate system is compared with its own matched global and S_0 baselines.

For state PCA32, the center and right singular vectors are fitted on the available training state-panel matrix only and applied unchanged to validation or test cells. The training and held-out embeddings therefore share one fitted coordinate frame. PCA coordinates are row-normalized before the primary prediction metrics.

5.10 Training-fitted target PCA32

The non-hash target analysis uses the fold-local target panel after target-only normalization. For each fit set, the target center μ_T and top 32 right singular vectors V_T are fitted only on fitting cells. Target coordinates are

$$Z_T^{\text{PCA}}(x) = (X_T(x) - \mu_T)V_T,$$

followed by row normalization for the primary metric. Singular values are ordered decreasingly, and the sign of each component is fixed by requiring its largest-absolute loading to be positive, with source-gene order breaking ties.

Each inner fit has its own target frame; each final outer frame excludes the held-out embryo. The global and S_0 baselines average normalized training target coordinates in the same fitted frame. A fixed unnormalized-PC sensitivity analysis uses the fitting-set mean squared PC norm as its error scale and avoids normalization of near-zero global means.

5.11 Nested whole-embryo model tuning and molecular prediction

For outer held-out embryo e , let

$$T_e = \{x : r(x) \neq e\}, \quad V_e = \{x : r(x) = e\}.$$

The training sizes are 2,672, 2,406, and 2,330 for held-out R1, R2, and R3. Every tunable model is selected before evaluating the outer test embryo.

Within the two outer-training embryos, each embryo is held out in turn as an inner validation fold. Gene eligibility, panel construction, representation fitting, and prediction are recomputed from the

single remaining inner-fit embryo. In that inner fit, the same eligibility rule is applied to the available fit set: detection fraction at least 1% in that one embryo, positive fit-owned mean expression, and the same mitochondrial/ribosomal exclusions. The inner-validation embryo contributes nothing to eligibility, panel assignment, normalization denominators, or representation fitting. Hyperparameters are chosen by minimum mean inner-fold NSE, with deterministic parameter-order tie breaking.

The molecular-only k -nearest-neighbor model uses squared Euclidean distance in the molecular representation and searches all training cells; no S_0 restriction is applied. Candidate neighborhood sizes are

$$k \in \{10, 25, 50\}.$$

For the selected k ,

$$\hat{Z}_T(x) = \frac{1}{k} \sum_{y \in N_k(x)} Z_T(y). \quad (20)$$

For the final inductive molecular-only and joint-metric models, all neighbor searches use the representation specified for that model after its stated normalization. Outer-test cells are excluded from the corresponding training search set, and inner-validation cells are excluded from the corresponding inner-fit search set. Distances are sorted with a full stable ascending sort, so equal-distance neighbors retain training-array order, which follows the frozen source-row order.

The preserved historical exact-key exposure calculation uses the original frozen 25-neighbor sets. Those sets were selected by `argpartition` at the 25th distance followed by a stable distance sort within the selected set. The numerical audit separately recomputes a full stable-sort 25-neighbor set and reports tie sensitivity, preserving the historical estimator while making the final-model tie convention explicit.

The global-mean baseline is the mean target vector over all outer-training cells. The S_0 -only baseline is the mean target vector among outer-training cells with matching S_0 ; when no matching cell exists, the training-global mean is used.

5.12 Prediction metrics

Target and prediction vectors are converted to float64 and rowwise L2-normalized before scoring. A vector with exactly zero L2 norm is divided by 1 and therefore remains the zero vector; no positive near-zero threshold is applied during normalization. The subsequent NSE denominator uses the explicit 10^{-12} floor shown below. Normalized squared error is

$$\text{NSE}_x = \frac{\|a_x - p_x\|_2^2}{\max(\|a_x\|_2^2, 10^{-12})}. \quad (21)$$

For nonzero unit vectors,

$$\text{NSE}_x = 2 - 2\langle a_x, p_x \rangle.$$

Cosine similarity and mean absolute coordinate error are also retained. Fold metrics are means over held-out cells. Biological conclusions are reported foldwise across the three embryos; cell-weighted aggregates are descriptive summaries only.

The molecular increment over coarse annotation in embryo e is

$$\Delta_{S,e} = \text{NSE}_{S_0,e} - \text{NSE}_{\text{molecular},e}. \quad (22)$$

5.13 Historical lineage-exposure audit

The historical estimator uses a fixed 25-cell molecular neighborhood and multiplies the contribution of exact history matches by two. For history level h , exposure is

$$m_h(x) = \#\{y \in N_{25}(x) : L_h(y) = L_h(x)\}. \quad (23)$$

The audit records the fractions with zero, one, and at least two matches, match-count summaries, the fraction of unchanged normalized weights, and total-variation distance between present-only and history-reweighted neighbor weights.

The historical foldwise gains are recomputed in float64 with deterministic neighbor and tie handling. Numerical stability is assessed separately from biological magnitude and foldwise replication.

5.14 Regularized and joint-metric lineage models

For continuous lineage models, the first six history coordinates are transformed by $\log(1 + x)$; all seven history coordinates are standardized using fitting cells only. Molecular predictors are standardized separately when required by the model family.

The regularized family compares molecular-only ridge prediction with ridge prediction from molecular state plus history. If X is the standardized fitting design, Y the target matrix, and $\bar{Y}$ the fitting target mean,

$$\hat{B}_\alpha = (X^\top X + \alpha I)^{-1} X^\top (Y - \bar{Y}), \quad (24)$$

and prediction restores the unpenalized target mean:

$$\hat{Y}(x) = \bar{Y} + x_{\text{std}}^\top \hat{B}_\alpha. \quad (25)$$

For history augmentation, $X_H = [X \mid H]$ after training-derived scaling,

$$B_{\alpha,H} = (X_H^\top X_H + \alpha I)^{-1} X_H^\top (Y - \bar{Y}),$$

with

$$\hat{Y}_H(x, h) = \bar{Y} + [x_{\text{std}}, h_{\text{std}}]^\top B_{\alpha,H}.$$

The original ridge grid is $\alpha \in \{0.01, 0.1, 1, 10\}$; the prospectively fixed capacity extension uses

$$\alpha \in \{0.01, 0.1, 1, 10, 100, 1000, 10000\}.$$

The joint-metric family uses

$$D_\lambda(x, y) = D_S(x, y) + \lambda D_H(x, y)/7, \quad (26)$$

with $k \in \{10, 25, 50\}$. The original metric grid uses $\lambda \in \{0, 0.01, 0.1, 1\}$; the extended grid uses

$$\lambda \in \{0, 0.0001, 0.001, 0.01, 0.1, 1, 10\}.$$

All hyperparameters are selected by the nested embryo procedure.

Within each model family, molecular-only and molecular-plus-history models are tuned independently using the same training-owned inner folds. Ridge penalties are selected separately for the molecular-only and molecular-plus-history designs. Likewise, molecular-only neighborhood size is selected independently of the joint-metric model’s neighborhood size and history-distance weight. Each lineage-augmented model is therefore compared with the independently selected molecular-only model from the corresponding family.

For each outer fold,

$$G_{H,e} = \text{NSE}_{\text{molecular},e} - \text{NSE}_{\text{molecular+history},e}. \quad (27)$$

A finite 3/3 directional result requires $G_{H,e} > 0$ for each embryo. A separate materiality indicator records whether every foldwise gain exceeds 10^{-4} .

5.15 Complete predictive lineage family and conditional calibration

The multiplicity unit is a configuration/model/stage comparison summarized over all three outer embryos. The 19 predictor/target configurations, two history-model comparisons, and two tuning stages give 76 three-embryo family members. Let

$$T_m = \min_e G_{m,e}, \quad T_{\max} = \max_{m=1,\dots,76} T_m.$$

A member statistic is positive only when its lineage-augmented model improves all three embryos.

For conditional calibration, molecular proximity is fixed without reference to target outcomes or lineage variables. A separate canonical state representation is constructed within each embryo from that embryo’s own eligible state-panel genes: genes must be detected in at least 1% of cells in that embryo, have positive mean expression, and pass the stated mitochondrial/ribosomal exclusions; state-panel counts use their own panel denominator, are projected with the canonical 64-dimensional state hash, and are rowwise L2-normalized.

Within each embryo and historical S_0 stratum, cells are then partitioned deterministically into molecular-neighborhood strata of at most 32 cells. Starting from the lowest remaining source-order cell, that cell is used as an anchor. Squared Euclidean distances from the anchor to all remaining cells in the same embryo- S_0 block are computed, and the anchor plus the nearest remaining cells are selected up to a total of 32. Distance ties are resolved by source-row index. The selected indices are removed, and the procedure repeats from the lowest remaining source-order cell until the block is exhausted. This target- and history-blind procedure produces 130 embryo- $\times$ S_0 $\times$ molecular-neighborhood strata, no singleton strata, a maximum stratum size of 32, and places all 3,704 cells in movable strata.

Within each null replicate, one random permutation is drawn independently within every stratum

and applied to the complete seven-variable history tuple

$$H = (d, p, g, e, ce, ba, ef).$$

The seven coordinates therefore move together and retain their observed joint support. The same permuted history assignment for a null seed is used throughout all history-bearing inner fits, tuning decisions, outer evaluations, model families, and predictor/target configurations entering that family-level null realization. Molecular predictors, target representations, S_0 , embryo membership, and held-out-fold ownership remain fixed.

Each of 999 fixed null seeds repeats all history-bearing inner-grid fits, two-inner-fold tuning, and outer evaluation. Molecular-only fits and tuning are invariant under history reassignment and are reused exactly. The family-wise probability is

$$p_{\text{FWER}} = \frac{1 + \#\{b : T_{\max}^{(b)} \geq T_{\max}^{\text{obs}}\}}{1000}.$$

Ties are inclusive and there is no early stopping. This finite-stratum calibration assumes exchangeability within the declared strata; it does not establish conditional independence given every continuous molecular coordinate.

5.16 Complete ambiguity/support diagnostics and simultaneous null

For each state-code size $K_S \in \{8, 16, 32, 64, 128\}$, target-code size K_T from the same set, and history level L_0 – L_4 , cells are grouped by the relevant state/history key. Within each key fiber F , let $n_{F,z}$ be the number of cells carrying target code z . Total collision support and target discordance are

$$P = \sum_F \binom{|F|}{2}, \quad D = \sum_F \left[\binom{|F|}{2} - \sum_z \binom{n_{F,z}}{2} \right],$$

with

$$A = D/P \tag{28}$$

when $P > 0$. Retained support at history level h is $U_h = P_h/P_0$.

The complete surface contains $5 \times 5 \times 5 = 125$ banks. A simultaneous conditional permutation family conditions on S_0 , embryo, and the joint vector of the five state codes. For each permutation the maximum observed ambiguity departure across the 100 non-baseline history contrasts is recorded. The resulting maximum-statistic null controls selection across the complete tested diagnostic family. Conditional permutations are finite diagnostics and are not treated as biological replication.

5.17 Continuation authority audit

A molecular ancestor-to-descendant continuation test requires a directly measured ancestral molecular state, a keyed lineage relation, and a measured descendant molecular outcome. The source reconstructed trees contain 2,686 internal nodes and zero directly measured internal-node

transcriptomes. Reconstructed edit state, inferred internal timing, and descendant summaries are not substituted for measured ancestral transcriptomes.

5.18 Verification and frozen authority

The final analysis specifications were frozen before their corresponding outcomes were computed. The preserved initial analysis remains unchanged.

Computational validation covers source identity and exact transcriptome–lineage joins, complete pair enumeration, matched-control construction, fold-local feature ownership, panel disjointness, panel-specific normalization, representation fitting, nested model selection, numerical replay, the complete predictor/target and diagnostic families, conditional permutations, finite-state sensitivity analyses, and figure/table consistency. The manuscript-specific public release provides the machine-readable validation manifest, numerical receipts, source hashes, realized analysis registries, and a standalone verifier that operates without access to the private development repository.

Formal controls additionally encode the principal logical distinctions used in the manuscript, including matched-null requirements for preferential history-resolution claims, predictor/target separation, estimator-scope limits for lineage conclusions, and exclusion of held-out embryos from fully inductive upstream construction. These controls verify computational invariants and stated logical implications; biological variable choice and scientific interpretation remain empirical modeling decisions.

6 Data availability

The primary data are MELA v1, Zenodo record 10.5281/zenodo.19892785, using the E7.5-R1, E7.5-R2, and E7.5-R3 source files.⁷ Raw third-party H5TD files remain at the official source archive. The Paper 2 reproducibility companion provides source hashes, annotation-provenance records, lineage-qualified population registries, fold-local gene vocabularies, panel registries, finite witness/control tables, prediction surfaces, history-model results, complete diagnostic banks, figure-data tables, and validation receipts.

7 Code availability

The public RMMO repository is available at

<https://github.com/zedjames/relational-morphogenesis>.

An isolated Paper 2 release package corresponding to this manuscript has been prepared and independently verified without a dependency on the private development repository. It contains analysis code, finite result tables, source manifests, validation receipts, formal controls, and

⁷Appendix A, Ref. R1.

manuscript/figure-data sources. The manuscript-specific package should be transferred to the public repository before external submission or acceptance; raw third-party MELA H5TD files remain at their official archive.

8 Author contributions

Zed James conceived the study, designed the relational and predictive analyses, developed the computational and formal framework, analyzed the results, and wrote the manuscript.

9 Competing interests

Competing interests: none declared.

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11 AI-assisted research and writing disclosure

Artificial intelligence tools were used as research-assistance software for code review, literature organization, manuscript drafting, and editorial revision. Scientific claims, source selection, computational authority, formal verification, interpretation, and final manuscript responsibility remain with the author.

A Reference notes

- R1.** Colgan, W. N., et al. (2026). *Comprehensive Lineage Tracing Maps the Landscape of Cell Fate Decisions in Mouse Embryogenesis*. bioRxiv. doi:10.64898/2026.05.07.722278. Data archive: MELA Zenodo v1, doi:10.5281/zenodo.19892785.
- R2.** Weinreb, C., Rodriguez-Fraticelli, A. E., Camargo, F. D., and Klein, A. M. (2020). *Lineage tracing on transcriptional landscapes links state to fate during differentiation*. Science 367, eaaw3381. doi:10.1126/science.aaw3381.
- R3.** Saelens, W., Cannoodt, R., Todorov, H., and Saeys, Y. (2019). *A comparison of single-cell trajectory inference methods*. Nature Biotechnology 37, 547–554. doi:10.1038/s41587-019-0071-9.
- R4.** Schiebinger, G., Shu, J., Tabaka, M., et al. (2019). *Optimal-transport analysis of single-cell gene expression identifies developmental trajectories in reprogramming*. Cell 176, 928–943.e22. doi:10.1016/j.cell.2019.01.006.
- R5.** Li, L., et al. (2023). *A mouse model with high clonal barcode diversity for joint lineage transcriptomic and epigenomic profiling in single cells*. Cell 186, 5183–5199.e22. doi:10.1016/j.cell.2023.09.019.
- R6.** Jia, Y., et al. (2026). *SPACE-seq integrates spatial transcriptomics and lineage tracing in native tissues*. Cell Stem Cell 33, 1398–1413.e10. doi:10.1016/j.stem.2026.05.017.
- R7.** Geiger, B. C., and Temmel, C. (2014). *Lumpings of Markov chains, entropy rate preservation, and higher-order lumpability*. Journal of Applied Probability 51, 1114–1132. doi:10.1239/jap/1421763331.
- R8.** Weinreb, C., and Klein, A. M. (2020). *Lineage reconstruction from clonal correlations*. Proceedings of the National Academy of Sciences 117, 17041–17048. doi:10.1073/pnas.2000238117.

B Previous analysis specification and motivation for the final inductive design

The initial analysis specification used all three embryos to define a 20,999-gene eligible vocabulary, deterministically split into 14,660 state genes and 6,339 target genes. Both panels were normalized by the complete 78,258-gene whole-cell library size. The initial predictor further restricted candidate neighbors to matching S_0 cells whenever at least 25 such training cells were available.

That historical estimator achieved cell-weighted NSE 0.090551 versus 0.102318 for its same-target training-global-mean reference, with improvement in all three held-out embryos. The historical exact-key lineage gains were

$$\begin{aligned} G_{L_1} &= -1.48847 \times 10^{-4}, & G_{L_2} &= -4.76443 \times 10^{-6}, \\ G_{L_3} &= +1.64276 \times 10^{-6}, & G_{L_4} &= +1.77429 \times 10^{-6}. \end{aligned}$$

with positive-fold counts 0/3, 1/3, 2/3, and 2/3.

These values remain valid descriptions of the frozen historical computation but can't remain the principal inferential results because whole-cell normalization allows target-panel counts to enter the predictor denominator, the candidate-neighbor rule uses S_0 , gene eligibility is transductively frozen across all embryos, and the deeper exact-history estimator has very low match exposure. The current manuscript retains these results to preserve analysis chronology and to make the transition to the fully inductive publication analysis auditable.

C Planned public reproducibility companion

The Paper 2 public companion is organized at:

[https://github.com/zedjames/relational-morphogenesis
papers/02-present-state-resolution-and-lineage-history/](https://github.com/zedjames/relational-morphogenesis/papers/02-present-state-resolution-and-lineage-history/)

The isolated release surface contains or is designed to contain source and sample authority, annotation-provenance records, fold-local gene vocabularies, state/target panel registries, signed-hash and PCA representation specifications, historical and disjoint-target witness/control tables, pair-bank decomposition, fully inductive prediction surfaces, lineage-exposure tables, regularized and joint-metric history results, all representation robustness tables, the complete 125-bank diagnostic surface, simultaneous permutation summaries, qualification-threshold sensitivity, numerical and formal verification receipts, figure-data sources, and manuscript hashes.

The release verifier is designed to operate on release-owned finite artifacts without access to the private development repository. Raw MELA H5TD files remain at the official source archive.